SHAO-CHUN HUNG

AI Engineer | Machine Learning | Computer Vision

+886-972-785-565 ✉ shaojun5861@gmail.com 🔗 Linkedin | GitHub 📍 Hsinchu, Taiwan

Summary

M.S. candidate in Mathematical Modeling and Scientific Computing at NYCU with hands-on experience in machine learning, computer vision, and AI system development. Built end-to-end ML pipelines spanning data preprocessing, model design, training, evaluation, error analysis, and research tooling, including cross-domain YOLO object detection and graph-based recommendation systems. Experienced with PyTorch, YOLO, OpenCV, Python, and AI-assisted engineering workflows.

Skills

Programming | Python · SQL · Matlab · JavaScript

Machine Learning | PyTorch · YOLO · Object Detection · Model Evaluation

Computer Vision | OpenCV · Data Augmentation · Image Processing

LLM / Agents | Harness Engineering · AI Agent Workflow

Engineering | Git · Linux · Docker · API Integration · CI/CD

Data | Pandas · NumPy · Data Visualization

Research & Projects

Cross-Domain Harbor–Airport Object Detection | YOLO11

NYCU

03/2025 - Present

- Built a **7-class YOLO11 detection pipeline** for cross-domain harbor and airport imagery.
- Developed a**nnotation/inference GUIs** and a **multi-view vessel-type assistant** for human review of missed and uncertain YOLO annotations
- Integrated **EfficientViM, DySample, and ASFF** to evaluate feature enhancement, dynamic upsampling, and multi-scale fusion.
- Audited and corrected dataset annotations, improving YOLO11n **mAP@50 from 76.2% to 77.0%** and demonstrating the impact of data quality on model performance.
- Designed **foreground compositing and harbor-domain data augmentation** to improve vehicle detection across multiple architectures.

Passenger Flow Analysis of the Taipei MRT

NTNU

10/2022 - 07/2023

- Built a **PageRank-based MRT recommender** using a **119×119 Markov transition matrix**, public ridership data, and the Power Method.
- Designed an **explainable Top-5 ranking algorithm** combining PageRank, OD connectivity, user preference, travel cost, and Min-Max normalization.
- Engineered robust asynchronous request handling with **AbortController, timeouts, retries, race-condition and stale-response protection**.
- Improved reliability and accessibility through **responsive testing, keyboard/IME support, SVG map interactions, regression tests, and security fixes**.

Awards

1st Place — 2026 TSMC IT CareerHack

- Developed an **AI-assisted legacy-code refactoring workflow** focused on reproducibility and verification.
- Designed **spec-driven prompts and validation workflows** to improve task alignment and output consistency.
- Built and validated workflows for **C-to-Rust migration and MVC architecture decoupling**, reducing the refactoring cycle time by **~80%.**

Best Research Award — 2022 Applied Mathematics Conference

- Led mathematical modeling, Python implementation, and result interpretation for the Taipei MRT passenger-flow project.
- Developed a recommendation approach to support tourist trip planning using interchange and passenger-flow patterns.

Experience

Teaching Assistant, **Calculus** | NYCU

Expected August 2026

- Developed an **AI-agent-assisted workflow** to support student registration and improve class administration efficiency.
- **Awarded Best Teaching Assistant of Academic Year 2025**

Education

National Yang Ming Chiao Tung University (NYCU)

Hsinchu, Taiwan

M.S. — Mathematical Modeling and Scientific Computing (Information Technology)

09/2024 - Present

National Taiwan Normal University (NTNU)

Taipei, Taiwan

B.S. — Mathematics

09/2020 - 06/2024

- GPA：**4.0/4.3** | Department Rank：**5/81**